

Home SIEM for Threat Detection – Portfolio Project

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1. Project Background

While studying cybersecurity and preparing for a career in Security Operations, I wanted to gain hands-on experience beyond theoretical knowledge.

To better understand how SOC analysts investigate security events, I built a Home SIEM lab using Splunk Enterprise, Windows Event Logs, and Sysmon.

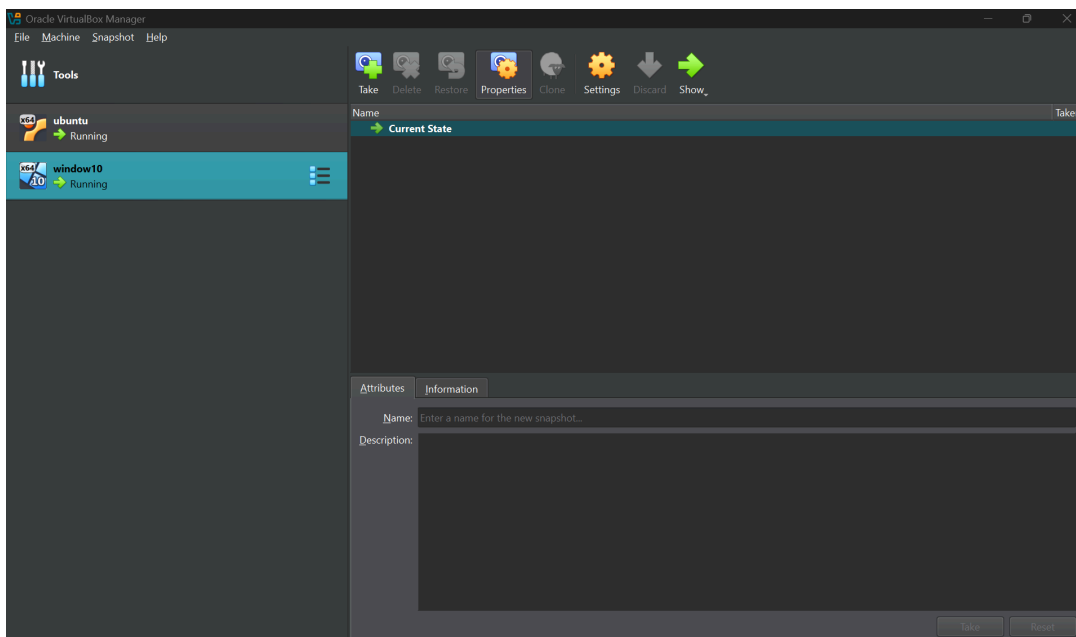
My goal was not only to learn how to configure security tools, but also to understand how SOC analysts think during an investigation. I wanted to practice identifying suspicious activity, validating hypotheses, documenting findings, and making decisions based on available evidence.

Throughout this project, I followed a workflow similar to that used by SOC analysts:

- Collecting and centralizing security telemetry
- Monitoring and reviewing security events
- Investigating suspicious activity
- Validating findings through log analysis
- Creating detections and alerts
- Documenting investigations and lessons learned
- Improving visibility through dashboards

This portfolio documents the environment I built, the investigations I performed, and the lessons learned throughout the process.

2. Building the Lab Environment



Why I Built This Environment

Before investigating security events, I needed an environment where I could safely simulate activity, collect telemetry, and practice the same investigative process used by SOC analysts in real-world environments.

To achieve this, I created two virtual machines using VirtualBox:

- A Windows 10 endpoint used to generate security events and simulate attacker activity.
- An Ubuntu Server running Splunk Enterprise to collect, index, and analyze logs.

Both systems were connected through an isolated host-only network, allowing me to safely test detections without affecting production systems.

Building the environment from scratch helped me understand how security telemetry flows from an endpoint to a SIEM platform and how analysts use that visibility to investigate suspicious activity and support security decisions.

3. SIEM Architecture & Log Pipeline

Why the Log Pipeline Was Important

Once the lab environment was operational, the next challenge was ensuring that security events generated on the Windows endpoint could be reliably collected and analyzed in Splunk.

Without centralized logging, investigating suspicious activity would be difficult because important evidence would remain scattered across individual systems.

To solve this, I designed a log pipeline that forwarded Windows Event Logs and Sysmon telemetry to Splunk Enterprise, allowing security events to be centrally monitored, investigated, and correlated

This setup mirrors a common workflow used in SOC environments where analysts rely on SIEM platforms to collect and correlate security events from multiple sources.

Building the Log Pipeline

Windows 10 Endpoint

The Windows endpoint acted as the primary source of security events throughout the project. It was used to simulate both normal user activity and suspicious behavior, providing realistic telemetry for investigation.

Sysmon

To gain deeper visibility into endpoint activity, I deployed Sysmon. This allowed me to capture detailed information about process execution, network connections, and registry changes that would later support multiple investigations.

Splunk Universal Forwarder

To ensure security events could be analyzed centrally, I configured Splunk Universal Forwarder to collect telemetry from the Windows endpoint and send it to the SIEM platform for investigation.

Splunk Enterprise

Splunk Enterprise served as the central investigation platform where logs were collected, searched, correlated, and visualized. This allowed me to review activity from multiple data sources and validate security findings.

Isolated Network

An isolated network was used to safely generate test activity without affecting production systems. This provided a controlled environment for validating detections and practicing investigative workflows.

How Logs Travel Through the Environment

Before beginning any investigation, I needed to confirm that security telemetry was being collected correctly. Reliable log collection was critical because every detection, alert, and investigation in the project depended on accurate data.

The following workflow illustrates how security events moved through the environment, from their creation on the endpoint to their analysis within the SIEM platform.

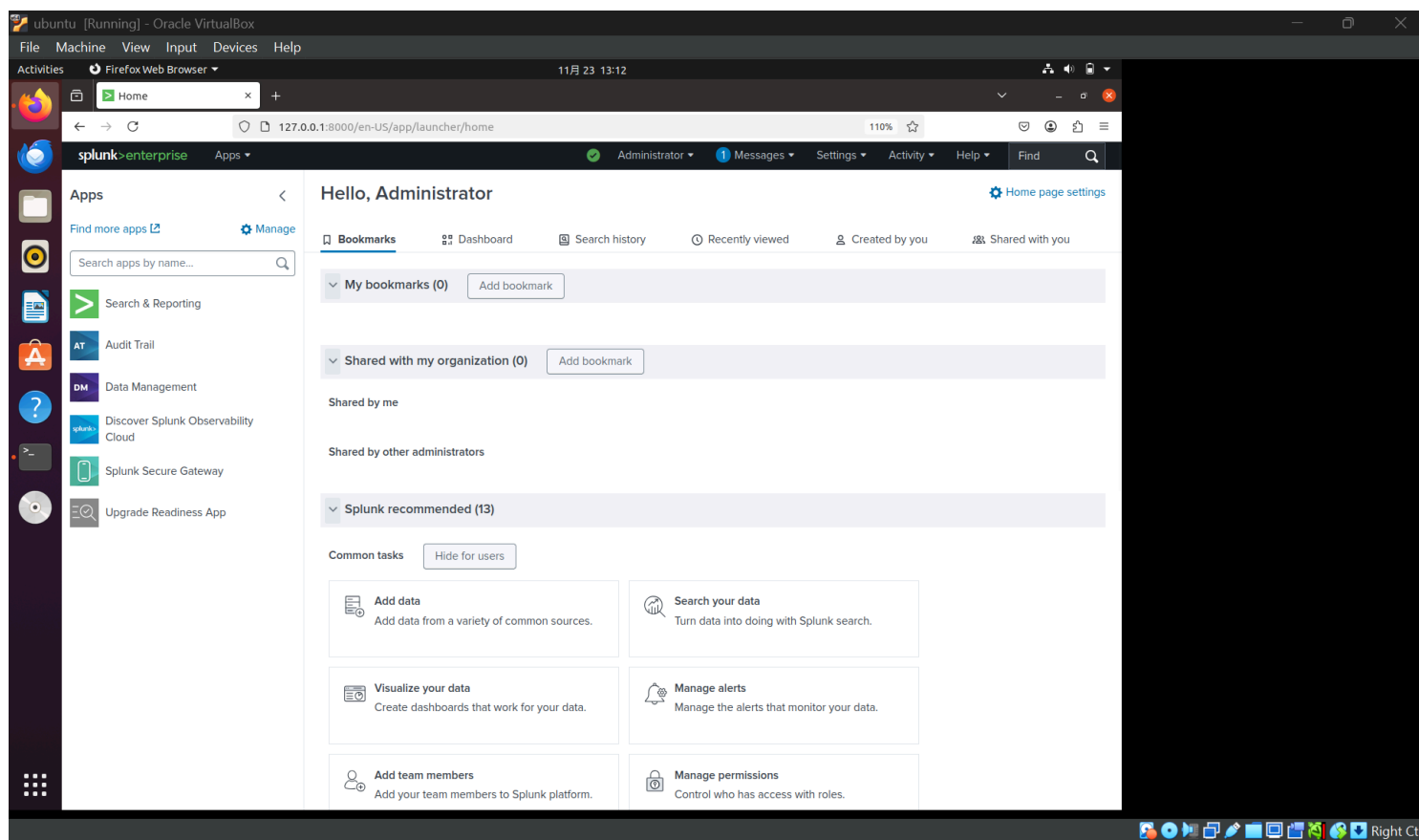
This validation step was important because every investigation performed later in the project depended on reliable log collection and visibility.

Windows 10 VM
(Log Source)
↓
Windows Event Logs + Sysmon
(Security, PowerShell, Sysmon Events)
↓
Splunk Universal Forwarder
(Log Collection & Forwarding)
↓
Ubuntu Server
(Splunk Enterprise SIEM)
↓
Splunk Searches, Alerts & Dashboards
(Threat Detection & SOC Triage)

Log Collection Process

1. Security activity was generated on the Windows endpoint through normal user actions and simulated attack scenarios.
2. Windows Event Logs and Sysmon recorded the activity and generated telemetry.
3. Splunk Universal Forwarder collected the telemetry and forwarded it to the SIEM platform.
4. Splunk Enterprise received and indexed the events.
5. The indexed data became available for searches, dashboards, detections, and investigations.

4. Validating the SIEM Platform



Confirming Visibility Before Investigation

Before relying on Splunk for investigations, I needed to confirm that the platform was successfully receiving and indexing security telemetry from the Windows endpoint.

I verified that the SIEM platform was operational, accessible, and capable of receiving data from the endpoint. This ensured that future investigations would be based on reliable telemetry.

This validation was important because every investigation, detection rule, dashboard, and alert in the project would depend on Splunk successfully collecting and indexing security events.

Establishing a reliable SIEM platform provided the visibility required to monitor endpoint activity, investigate suspicious events, validate detections, and support security investigations throughout the project.

5. Verifying End-to-End Log Collection

Preparing for Log Collection

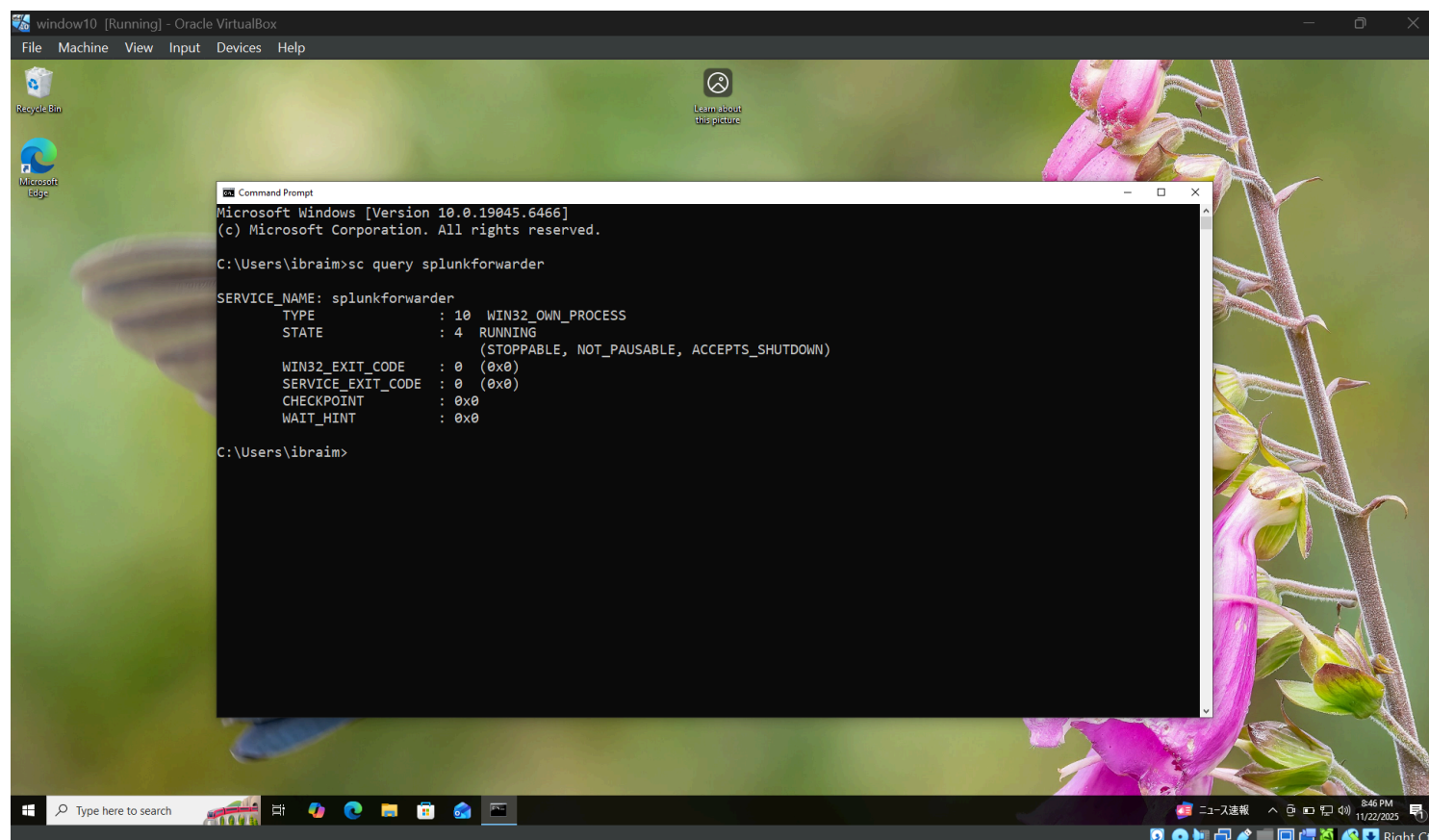
After confirming that Splunk Enterprise was operational, I needed to ensure that security logs generated on the Windows endpoint could successfully reach the SIEM platform.

To achieve this, I verified that the Splunk Universal Forwarder service was running correctly on the Windows machine and actively collecting telemetry from Windows Event Logs and Sysmon.

This step was critical because any interruption in log forwarding would prevent security events from reaching Splunk, reducing visibility and impacting future investigations.

Verifying the Universal Forwarder

Before conducting any security investigation, I verified that Splunk Universal Forwarder was actively collecting and forwarding telemetry from the Windows endpoint. This ensured that security events would be available for monitoring, detection, investigation and analysis within the SIEM platform.



Why This Validation Matters

The Splunk Universal Forwarder acts as the bridge between the Windows endpoint and the SIEM platform.

By confirming that the service was running successfully, I verified that security events could be collected and transmitted to Splunk for analysis.

This validation provided confidence that the log pipeline was functioning as expected and that future detections and investigations would be based on complete and reliable telemetry.

Confirming Log Visibility in Splunk

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 11月 23 14:16

Search | Splunk 10.0.1

splunk>enterprise Apps Administrator Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search Save As Create Table View Close

index=* host="DESKTOP-BR4IA80" Time range: Last 60 minutes

✓ 141 events (11/23/25 1:14:00.000 PM to 11/23/25 2:14:58.000 PM) No Event Sampling Job II Smart Mode

Events (141) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 3
- sourcetype 3

INTERESTING FIELDS

- Account_Domain 3
- Account_Name 4
- ComputerName 1
- EventCode 26
- EventType 4
- index 1
- Keywords 6
- linecount 11
- LogName 3
- Logon_ID 3
- Message 39
- OpCode 5
- punct 31
- Read_Operation 1

i	Time	Event
>	11/23/25 2:14:29.424 PM	11/22/2025 09:14:29.424 PM LogName=System EventCode=134 EventType=3 ComputerName=DESKTOP-BR4IA80 Show all 15 lines host = DESKTOP-BR4IA80 source = WinEventLog:System sourcetype = WinEventLog:System
>	11/23/25 2:14:27.914 PM	11/22/2025 09:14:27.914 PM LogName=System EventCode=134 EventType=3 ComputerName=DESKTOP-BR4IA80 Show all 15 lines host = DESKTOP-BR4IA80 source = WinEventLog:System sourcetype = WinEventLog:System
>	11/23/25 2:14:27.898 PM	11/22/2025 09:14:27.898 PM LogName=System EventCode=134 EventType=3 ComputerName=DESKTOP-BR4IA80

Confirming Log Visibility in Splunk

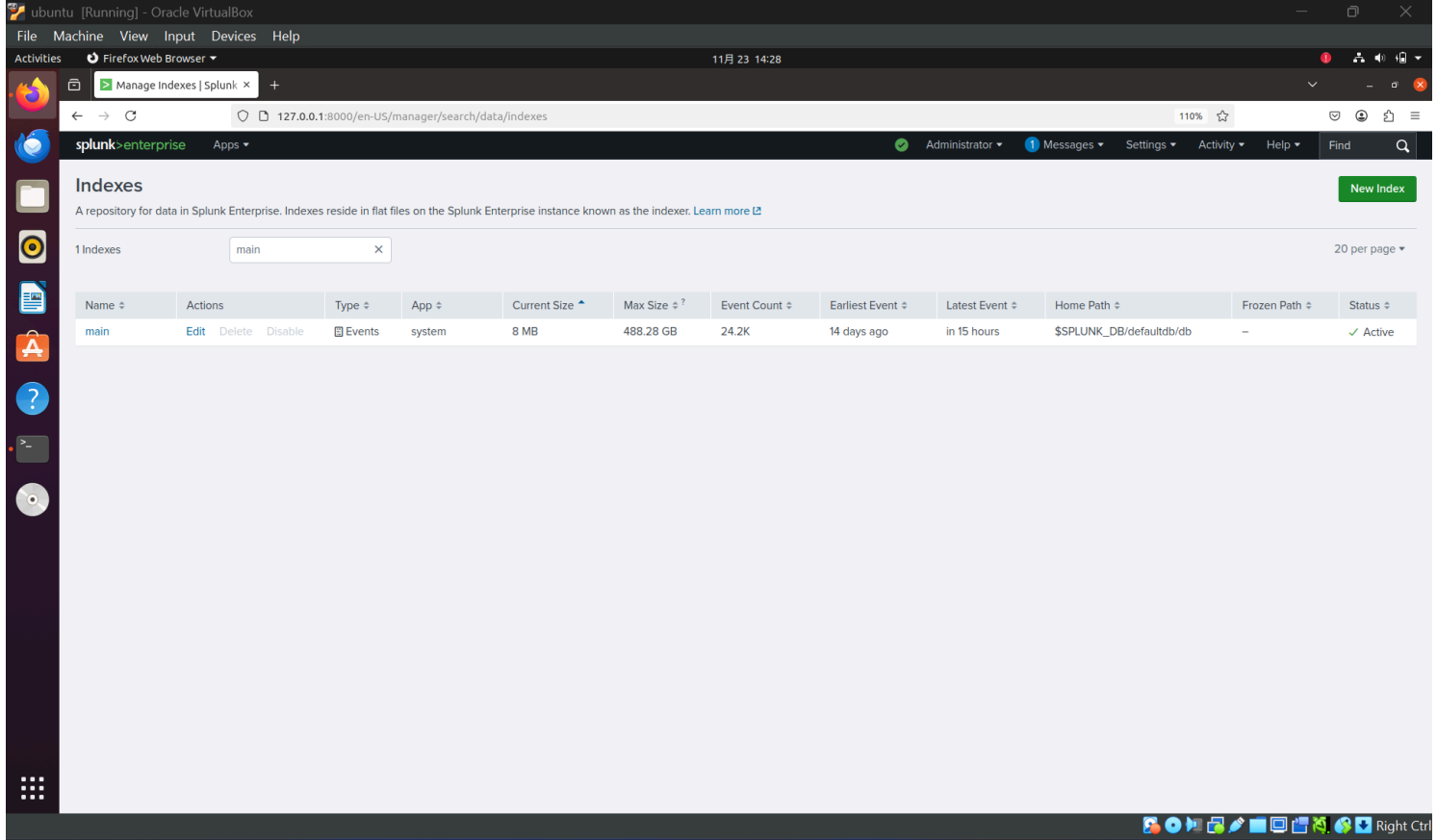
After validating the log forwarding process, I confirmed that security events generated on the Windows endpoint were successfully reaching and being indexed by Splunk.

I reviewed Windows Event Logs and Sysmon telemetry within Splunk to verify that endpoint activity was visible and available for investigation.

At this stage, I confirmed that Splunk was receiving and indexing events generated on the Windows endpoint, providing the visibility required for threat detection and investigation activities.

This validation marked the point where the SIEM environment became fully operational and ready for security monitoring.

Verifying Continuous Log Ingestion



Verifying Continuous Log Ingestion

Successful monitoring depends on continuous visibility, not just the receipt of a few isolated events.

To confirm the stability of the environment, I monitored log ingestion over time and verified that events continued to arrive in the Splunk index without interruption.

This validation confirmed that the forwarding pipeline remained active and that security telemetry could be continuously collected for future detections, investigations, and dashboard monitoring.

Establishing reliable log ingestion was essential before moving on to creating custom detection rules.

Investigation 1 – Brute Force Login Activity

Situation

While monitoring security events in Splunk, I wanted to determine whether repeated failed authentication attempts could be detected and investigated using Windows Security logs.

Since brute-force attacks commonly generate multiple failed logon events, I decided to simulate this activity within the lab environment and validate the visibility provided by the SIEM platform.

Initial Hypothesis

The failed login attempts could represent normal user behavior, such as repeatedly entering an incorrect password. However, a high volume of authentication failures occurring within a short period may also indicate password-guessing or brute-force activity.

My objective was to determine whether the observed behavior represented a benign user mistake or a pattern that would require further investigation by a SOC analyst.

Analyst Thinking

At this stage, I considered two possible scenarios:

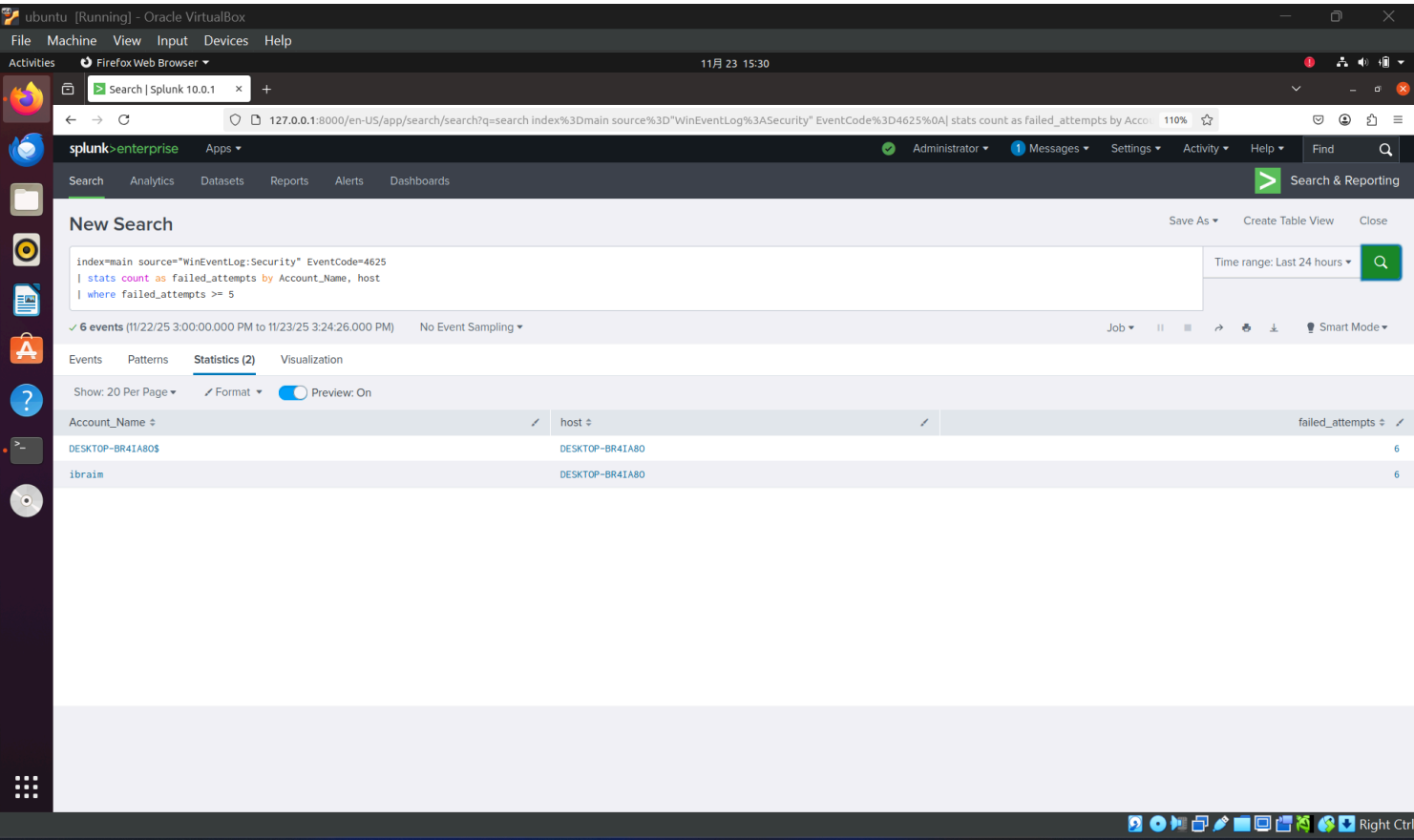
- A legitimate user repeatedly entering an incorrect password.
- An attacker attempting to gain access through password guessing.

To differentiate between these possibilities, I needed to review authentication frequency, affected accounts, and event timing to determine whether the activity appeared suspicious.

Investigation

To investigate the activity, I reviewed Windows Security Event ID 4625 logs in Splunk and created a correlation search designed to identify accounts generating multiple failed authentication attempts within a short period.

By grouping events by account name and host, I was able to identify authentication patterns that would typically require triage and validation by a SOC analyst.



Findings

The analysis revealed multiple failed authentication attempts associated with the monitored Windows endpoint. The correlation search successfully identified accounts exceeding the configured threshold, demonstrating how repeated authentication failures can be detected through centralized log analysis. Although the activity was generated within a controlled lab environment, the pattern closely resembled behavior commonly associated with brute-force attempts.

Response

After validating the activity, I created a Splunk detection rule designed to generate an alert whenever failed authentication attempts exceeded a predefined threshold. This improved visibility into suspicious authentication activity and reduced the time required to identify potential brute-force behavior.

In a production environment, this alert would support rapid triage and investigation by the SOC team.

Escalation Considerations

In a production environment, I would consider escalating this activity to a Tier 2 analyst if:

- The failed login attempts targeted privileged or administrative accounts.
- Multiple user accounts were targeted from the same source.

- Similar authentication failures were observed across multiple hosts.
- A successful login occurred after repeated failed authentication attempts.
- The activity was accompanied by other suspicious events, such as PowerShell execution or unusual process activity.

These indicators could increase the likelihood of malicious activity and would require deeper investigation to determine the scope and potential impact of the event.

MITRE ATT&CK Mapping

T1110 – Brute Force

Lessons Learned

This investigation reinforced the importance of correlating authentication events over time rather than reviewing individual failed logins in isolation.

Even in a small lab environment, repeated failed authentication attempts can provide valuable indicators of suspicious activity that may warrant further investigation.

Investigation 2 – Suspicious PowerShell Activity

Situation

While reviewing endpoint telemetry in Splunk, I wanted to determine whether PowerShell activity could be monitored effectively and whether potentially suspicious commands could be identified through centralized logging.

Since PowerShell is commonly abused by attackers for reconnaissance, payload execution, and system administration tasks, visibility into script execution was an important capability to validate within the lab environment.

Initial Hypothesis

Not all PowerShell activity is malicious. Many commands are executed as part of normal administrative operations.

However, unusual or unexpected PowerShell commands may indicate malicious behavior and should be investigated to determine their intent and impact.

Analyst Thinking

When I first reviewed the PowerShell activity, I did not immediately assume the commands were malicious.

PowerShell is commonly used by administrators for legitimate tasks, but it is also frequently abused by attackers for reconnaissance, execution, persistence, and lateral movement.

My goal was to determine whether the observed commands represented normal administrative activity or behavior that would warrant further investigation in a SOC environment.

Investigation

To investigate PowerShell activity, I analyzed Windows PowerShell Operational logs (Event ID 4104) collected in Splunk.

The search focused on PowerShell script block logging, allowing me to review executed commands and identify activity that would warrant further investigation in a real SOC environment.

I reviewed the recorded commands to understand what actions were performed on the endpoint and to determine whether any behavior appeared unusual or potentially suspicious.

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 11月 23 19:52

Splunk | The Key to Enter: Search | Splunk 10.0.1

New Search

index=main sourcetype="XmlWinEventLog:Microsoft-Windows-PowerShell/Operational" EventID=4104
| table _time host ScriptBlockText
| sort -_time

Time range: Last 7 days

103 events (11/16/25 7:00:00.000 PM to 11/23/25 7:52:03.000 PM) No Event Sampling

Events Patterns Statistics (103) Visualization

Show: 20 Per Page Format Preview: On

_time	host	ScriptBlockText
2025-11-23 19:38:17.069	DESKTOP-BR41A80	prompt
2025-11-23 19:38:17.062	DESKTOP-BR41A80	echo final-ok-2
2025-11-23 19:38:17.031	DESKTOP-BR41A80	echo final-ok-1 IEX "echo final-ok-2"
2025-11-23 17:22:28.493	DESKTOP-BR41A80	prompt
2025-11-23 17:22:28.434	DESKTOP-BR41A80	echo test-pro2
2025-11-23 17:22:28.385	DESKTOP-BR41A80	echo test-pro IEX "echo test-pro2"
2025-11-23 17:11:17.579	DESKTOP-BR41A80	prompt
2025-11-23 17:11:17.574	DESKTOP-BR41A80	echo test123
2025-11-23 17:11:17.568	DESKTOP-BR41A80	echo suspicious IEX "echo test123"
2025-11-23 17:11:13.871	DESKTOP-BR41A80	prompt
2025-11-23 17:11:13.851	DESKTOP-BR41A80	prompt
2025-11-23 17:11:00.618	DESKTOP-BR41A80	cd "C:\Program Files\SplunkUniversalForwarder\bin" .\splunk restart
2025-11-23 17:10:16.079	DESKTOP-BR41A80	prompt
2025-11-23 17:10:16.048	DESKTOP-BR41A80	notepad "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf"
2025-11-23 17:09:14.038	DESKTOP-BR41A80	prompt

Findings

The PowerShell logs provided detailed visibility into commands executed on the Windows endpoint and confirmed that script block logging was functioning correctly.

Several administrative and test commands were successfully captured, demonstrating that PowerShell activity could be monitored and reviewed through centralized logging.

The investigation confirmed that command execution details were available for analysis, allowing suspicious PowerShell activity to be identified and investigated more effectively.

Response

To improve visibility into endpoint activity, I incorporated PowerShell logging into the overall monitoring strategy and validated that Event ID 4104 telemetry was consistently reaching the SIEM platform.

This ensured that future investigations involving PowerShell activity could be supported with detailed command execution data, improving visibility and reducing investigation time.

Escalation Considerations

In a production environment, I would consider escalating PowerShell activity if:

- Encoded or obfuscated commands were observed.
- PowerShell was used to download or execute external content.
- Commands attempted credential access or privilege escalation.
- The activity originated from unexpected users or systems.
- The PowerShell execution was associated with other suspicious events.

These indicators could suggest malicious use of PowerShell and would require deeper investigation to determine the scope and impact of the activity.

MITRE ATT&CK Mapping

T1059.001 – PowerShell

Lessons Learned

This investigation highlighted the importance of PowerShell logging as a source of endpoint visibility.

Having access to script block logs provides significantly greater insight than relying solely on standard Windows Event Logs and can help analysts identify potentially suspicious behavior earlier in the investigation process.

The exercise also reinforced the importance of validating context before classifying PowerShell activity as malicious, since many legitimate administrative tasks use the same tool.

Investigation 3 – Monitoring Process Creation Activity

Situation

After validating authentication and PowerShell visibility, I wanted to improve endpoint monitoring by reviewing process creation activity captured through Sysmon Event ID 1.

Process creation telemetry provides valuable insight into what is executing on a system and can help identify suspicious or unauthorized activity.

Initial Hypothesis

Most process creation events are generated by normal operating system activity or legitimate user actions.

However, unusual process executions, unexpected command-line arguments, or suspicious parent-child process relationships may indicate malicious behavior and should be investigated further.

Analyst Thinking

When I first reviewed the process creation events, I did not immediately assume that any process was malicious.

Most process execution activity is generated by normal operating system functions or legitimate user actions. However, unusual process names, unexpected command-line arguments, or suspicious parent-child process relationships can indicate potentially malicious behavior.

My objective was to determine whether the observed activity reflected normal system operations or whether it contained indicators that would warrant further investigation in a SOC environment.

Investigation

To investigate process execution activity, I analyzed Sysmon Event ID 1 logs collected in Splunk.

The search provided visibility into executed processes, command-line arguments, parent processes, and associated user accounts.

I reviewed the collected telemetry to understand what applications were running on the endpoint and to identify any execution patterns that appeared unusual or potentially suspicious.

ubuntu [Running] - Oracle VirtualBox

Activities Machine View Input Devices Help

Firefox Web Browser 11月 23 22:47

Restore Session Search | Splunk 10.0.1

127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3D"XmlWinEventLog%3AMicrosoft-Windows-Sysmon%2FOperational" EventID%3D1%0A| 110%

splunk-enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventID=1
| table _time host Image CommandLine User
| sort -_time
```

Time range: Last 24 hours

✓ 1,024 events (11/22/25 10:00:00.000 PM to 11/23/25 10:45:20.000 PM) No Event Sampling Job Smart Mode

Events Patterns Statistics (1,024) Visualization

Show: 20 Per Page Format Preview: On

_time	host	Image	CommandLine	User
2025-11-23 22:34:05.734	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:34:04.578	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	NT AUTHORITY\SYSTEM
2025-11-23 22:34:03.422	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:33:57.863	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:33:55.346	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:33:54.201	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:32:39.403	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:32:34.906	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	NT AUTHORITY\SYSTEM
2025-11-23 22:32:33.400	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:32:27.865	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:32:25.348	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:32:24.198	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:31:52.630	DESKTOP-BR4IA80	C:\Windows\System32\MoUsocoreWorker.exe	C:\Windows\System32\mousocoreworker.exe -Embedding	NT AUTHORITY\SYSTEM
2025-11-23 22:31:04.942	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"	NT AUTHORITY\SYSTEM
2025-11-23 22:31:03.209	DESKTOP-BR4IA80	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	NT AUTHORITY\SYSTEM

Findings

The collected Sysmon telemetry successfully captured process creation events generated on the Windows endpoint.

The logs provided detailed visibility into executed applications, command-line activity, and user context, demonstrating that endpoint process monitoring was functioning correctly within the SIEM environment.

The investigation confirmed that process execution activity could be reviewed and analyzed effectively, providing valuable context for future security investigations.

Response

To improve endpoint visibility, I incorporated Sysmon Event ID 1 telemetry into the overall monitoring strategy and validated that process creation events were consistently reaching Splunk.

This provided greater visibility into executed applications, command-line activity, and user context, supporting faster analysis and investigation of endpoint activity.

Escalation Considerations

In a production environment, I would consider escalating process execution activity if:

- Suspicious or unknown processes were executed.
- Unusual command-line arguments were observed.
- Unexpected parent-child process relationships were identified.
- System administration tools were executed by non-administrative users.
- Process execution was associated with other suspicious events or alerts.

These indicators could suggest malicious execution activity and would require deeper investigation to determine the scope and potential impact of the event.

MITRE ATT&CK

T1059 – Command and Scripting Interpreter

Lessons Learned

This investigation demonstrated the value of process creation telemetry during endpoint investigations.

Having visibility into executed processes, command-line arguments, and parent-child relationships provides analysts with important context when investigating suspicious activity.

The exercise also reinforced how Sysmon Event ID 1 can help identify potentially malicious behavior that may not be visible through standard Windows logging alone.

Investigation 4 – Registry Persistence Activity

Situation

While reviewing Sysmon telemetry, I wanted to determine whether registry modifications associated with persistence techniques could be detected and investigated through centralized logging.

Persistence mechanisms are commonly used by attackers to maintain access to a compromised system, making visibility into registry activity an important capability for security monitoring.

Initial Hypothesis

Not all registry modifications are malicious. Many legitimate applications modify registry values during installation, updates, or normal system operation.

However, changes to autorun locations such as Run and RunOnce registry keys may also indicate an attempt to establish persistence and should be investigated further.

Analyst Thinking

When I first reviewed the registry modification events, I did not immediately assume the activity was malicious.

Registry changes occur frequently during normal system operation, software installation, and application updates. However, modifications involving autorun locations are commonly associated with persistence techniques used by attackers to maintain access to a compromised system.

My objective was to determine whether the observed registry activity represented legitimate system behavior or changes that would warrant further investigation in a SOC environment.

Investigation

To investigate registry activity, I analyzed Sysmon Event ID 13 logs collected in Splunk.

The search focused on registry value modifications and provided visibility into affected registry paths, modified values, and associated user accounts.

I reviewed the collected telemetry to determine what changes were made, which registry locations were affected, and whether any modifications were consistent with persistence-related behavior commonly observed in real-world attacks.

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 11月 24 21:08

Search | Splunk 10.0.1

127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3D"XmlWinEventLog%3AMicrosoft-Windows-Sysmon%2FOperational" EventID%3D13%0A 110%

splunk>enterprise Apps Administrator Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search Save As Create Table View Close

index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventID=13
| table _time host TargetObject Details User
| sort -_time

Time range: Last 15 minutes

334 events (11/24/25 8:53:38.000 PM to 11/24/25 9:08:38.000 PM) No Event Sampling Job

Events Patterns Statistics (334) Visualization

Show: 20 Per Page Format Preview: On

_time	host	TargetObject	Details	User
2025-11-24 21:01:49.102	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.102	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.101	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.100	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.098	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.021	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.021	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.021	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.018	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.015	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.015	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.010	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:49.006	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:48.754	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM
2025-11-24 21:01:48.753	DESKTOP-BR4IA80			NT AUTHORITY\SYSTEM

Findings

The collected Sysmon telemetry successfully captured registry modification events generated on the Windows endpoint.

The logs provided visibility into registry paths, modified values, and associated user activity, demonstrating that persistence-related events could be detected and reviewed through Splunk.

The investigation confirmed that registry modification activity could be monitored effectively and used to identify changes that may indicate persistence techniques.

Response

To improve visibility into persistence-related activity, I incorporated Sysmon Event ID 13 telemetry into the overall monitoring strategy and validated that registry modification events were consistently reaching the SIEM platform.

This provided greater visibility into registry-based persistence techniques and supported faster investigation of potentially suspicious system changes.

Escalation Considerations

In a production environment, I would consider escalating registry modification activity if:

- Changes were made to autorun registry locations such as Run or RunOnce.
- Registry modifications were associated with suspicious or unknown processes.
- Multiple persistence-related changes were observed within a short period.
- Registry activity was linked to other suspicious events or alerts.
- The modifications involved privileged accounts or critical system locations.

These indicators could suggest an attempt to establish persistence on a system and would require deeper investigation to determine the scope and potential impact of the activity.

MITRE ATT&CK

T1547.001 – Registry Run Keys / Startup Folder

Lessons Learned

This investigation highlighted the importance of monitoring registry activity as part of an endpoint detection strategy.

Registry modifications can provide valuable indicators of persistence techniques and help analysts identify suspicious activity before malicious processes are executed.

The exercise also reinforced the value of Sysmon Event ID 13 as a source of visibility into system changes that may otherwise go unnoticed through standard Windows logging.

6.5 MITRE ATT&CK Mapping

Throughout this project, each investigation was mapped to relevant MITRE ATT&CK techniques to better understand how security events relate to real-world adversary behavior.

The table below summarizes the techniques identified during the investigations performed in the lab.

Detection Use Case	MITRE ATT&CK Technique	Technique ID
Brute-Force Login Attempts	Brute Force	T1110
Suspicious PowerShell Execution	Command and Scripting Interpreter: PowerShell	T1059.001
Process Creation Monitoring	Process Discovery	T1057
Registry Persistence Detection	Boot or Logon Autostart Execution: Registry Run Keys	T1547.001

7. Attack Simulations

Simulation 1 – Brute Force Login Activity

Objective

To generate failed authentication events and validate that Event ID 4625 telemetry was successfully collected and indexed by the SIEM platform.

Attack Simulation

Multiple failed login attempts were intentionally generated by entering incorrect credentials several times on the Windows endpoint.

ubuntu [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

ActivitiesFirefox Web Browser

11月 24 19:42

127.0.0.1:8000/en-US/app/search/search?earliest=-15m&latest=now&q=search index%3Dmain source%3D"WinEventLog%3A\Security" EventCode%3D4625%0A| table _time host Account_Name Logon_Type Failure_Reason | sort -_time

splunk>enterprise

AdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

Search & Reporting

New Search

Save AsCreate Table ViewClose

index=main source="WinEventLog:Security" EventCode=4625
| table _time host Account_Name Logon_Type Failure_Reason
| sort -_time

Time range: Last 15 minutes

8 events (11/24/25 7:26:45.000 PM to 11/24/25 7:41:45.000 PM) No Event Sampling

Job

Smart Mode

EventsPatternsStatistics (8)Visualization

Show: 20 Per PageFormatPreview: On

_time	host	Account_Name	Logon_Type	Failure_Reason
2025-11-24 19:38:21.346	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:37:53.730	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:37:10.699	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:37:06.987	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:37:03.012	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:36:58.489	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:36:54.955	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.
2025-11-24 19:36:50.782	DESKTOP-BR4IA80	DESKTOP-BR4IA80\$ibraim	2	Unknown user name or bad password.

127.0.0.1:8000/en-US/app/search/search?earliest=-15m&latest=now&q=search index=main source="WinEventLog:Security" EventCode=4625...search.mode=smart&dispatch.sample_ratio=1&workload_pool=&display.page.search.tab=statistics&display.general.type=statistics#

Right Ctr

Validation Result

The generated Event ID 4625 events were successfully forwarded to Splunk, indexed correctly, and made available for investigation through centralized logging.

Outcome

This simulation confirmed that failed authentication activity could be detected, monitored, and investigated using the SIEM platform.

Simulation 2 – Suspicious PowerShell Activity

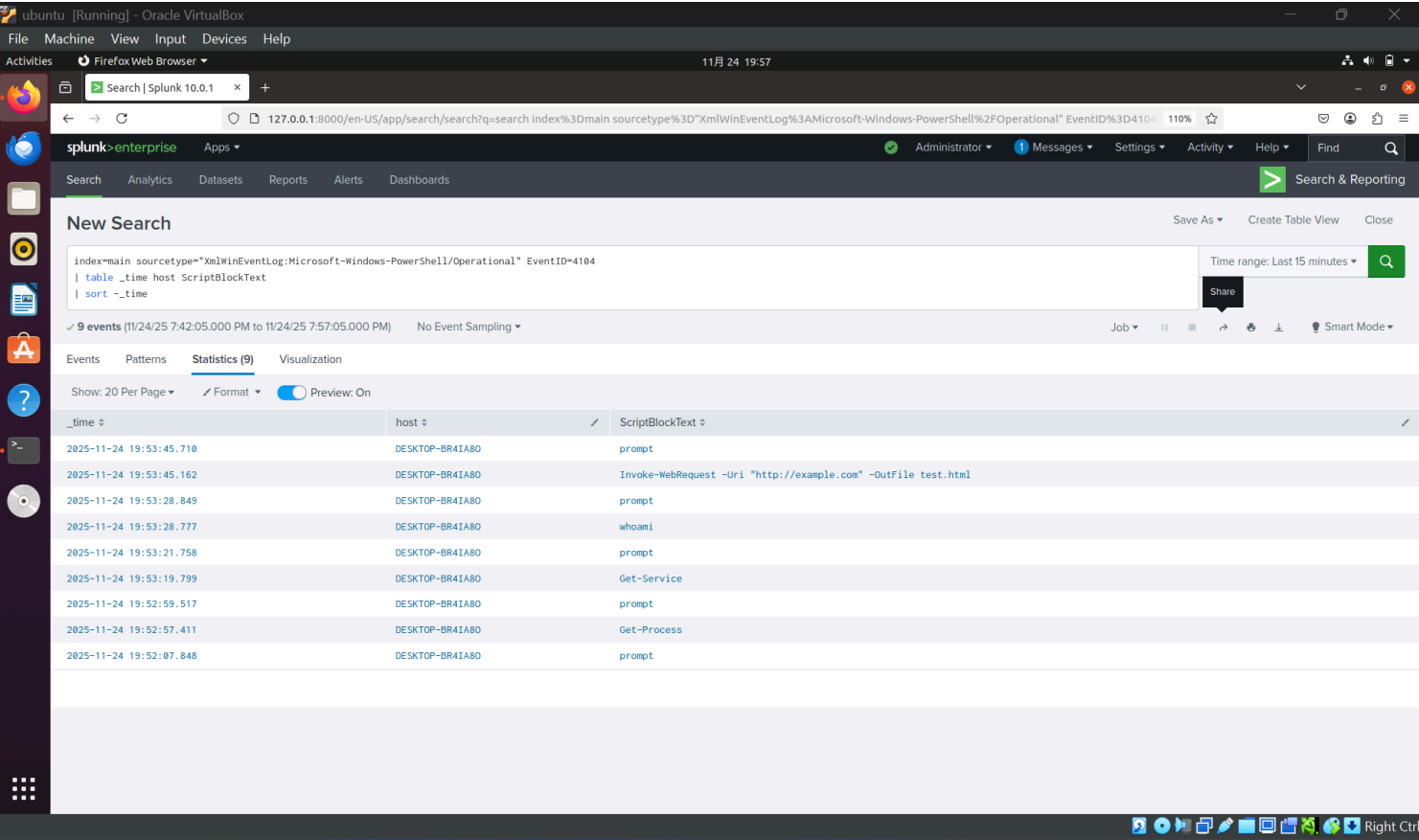
Objective

To generate PowerShell activity and validate that Event ID 4104 telemetry was successfully collected and searchable within Splunk.

Attack Simulation

Several PowerShell commands were intentionally executed on the Windows endpoint, including common administrative commands and a simulated suspicious command using Invoke-WebRequest.

The objective was to generate Event ID 4104 entries and verify that detailed PowerShell activity was successfully forwarded to Splunk.



Validation Result

The executed commands were successfully captured and indexed, providing visibility into PowerShell activity through centralized logging.

Outcome

This simulation confirmed that PowerShell execution activity could be monitored and investigated through Splunk, providing the visibility required to identify suspicious commands and support security investigations.

Simulation 3 – Process Creation Activity

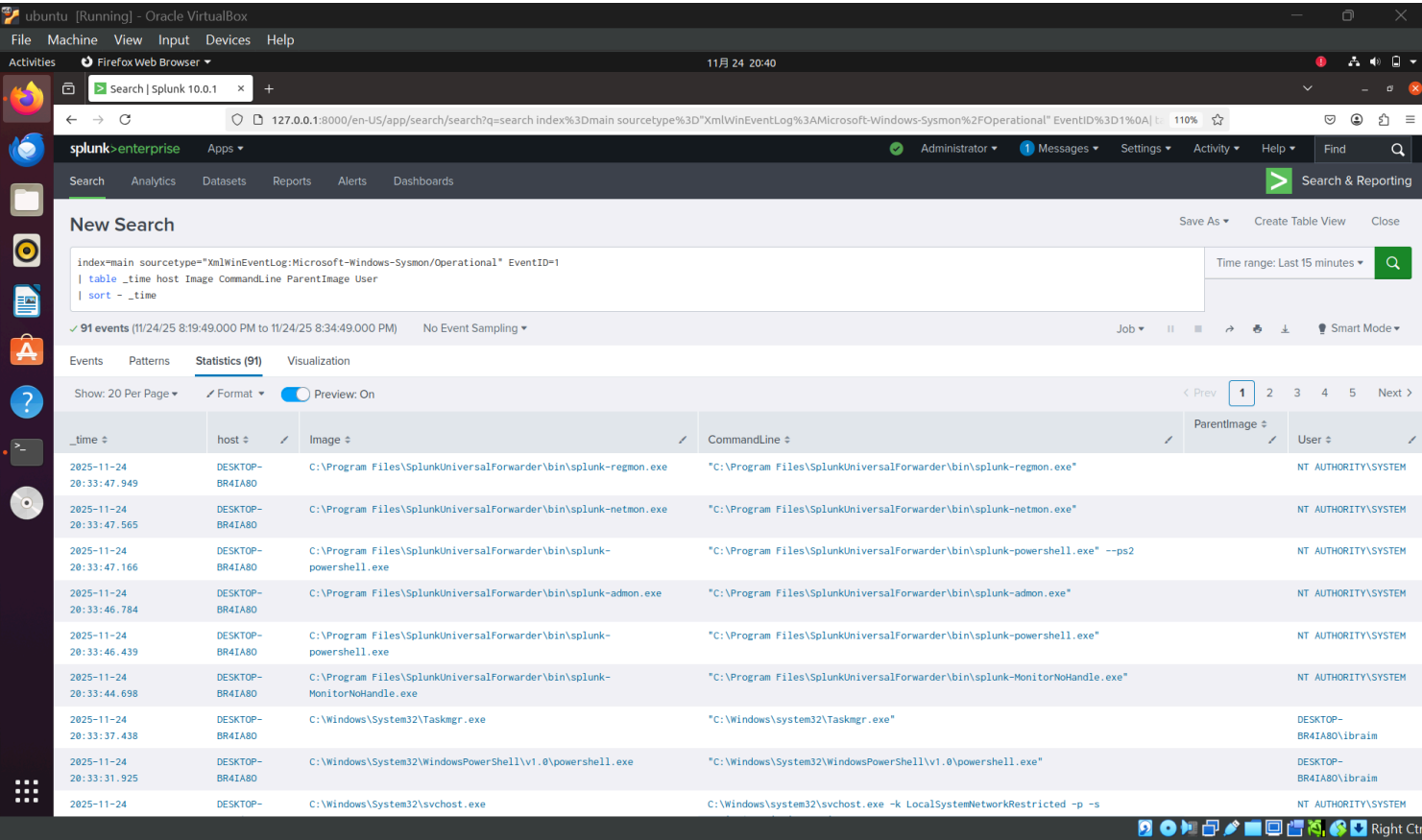
Objective

To validate that Sysmon Event ID 1 telemetry could be successfully collected and analyzed through Splunk.

Attack Simulation

Several Windows applications were intentionally executed on the endpoint, including Notepad, Calculator, Task Manager, and PowerShell.

The objective was to generate Sysmon Event ID 1 process creation events and verify that endpoint activity was successfully forwarded to Splunk for monitoring and investigation.



Validation Results

The simulation successfully generated Sysmon Event ID 1 telemetry and provided detailed visibility into process execution activity.

Outcome

This simulation confirmed that process creation telemetry could be monitored and investigated through Splunk, providing visibility into executed applications, command-line activity, parent-child process relationships, and user context.

This information can help analysts identify suspicious execution patterns, investigate endpoint activity, and better understand how processes interact within a system.

Simulation 4 – Registry Persistence Activity

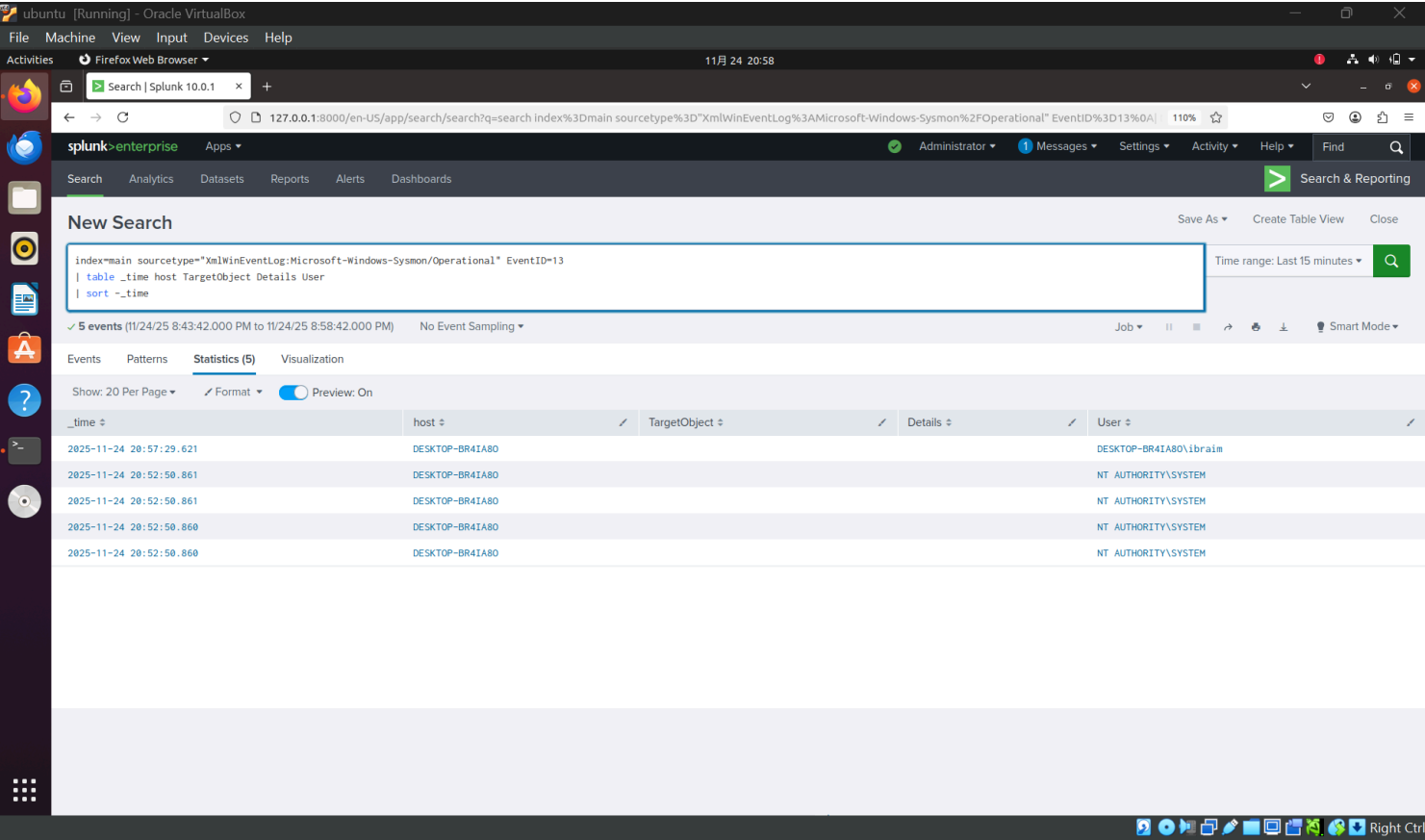
Objective

To validate that Sysmon Event ID 13 telemetry could successfully detect and record registry modifications associated with persistence techniques.

Attack Simulation

A registry Run key entry was intentionally created within the Windows endpoint to simulate a persistence mechanism commonly used by attackers.

The objective was to generate Sysmon Event ID 13 events and verify that registry modification activity was successfully forwarded to Splunk for monitoring and investigation.



Validation Results

The simulation successfully generated Sysmon Event ID 13 telemetry and provided visibility into registry modification activity.

The collected events included affected registry paths, modified values, and associated user accounts, demonstrating that persistence-related registry activity was successfully collected, indexed, and searchable within Splunk.

Outcome

This simulation confirmed that registry modification activity could be monitored and investigated through Splunk, providing visibility into persistence-related changes made on the endpoint.

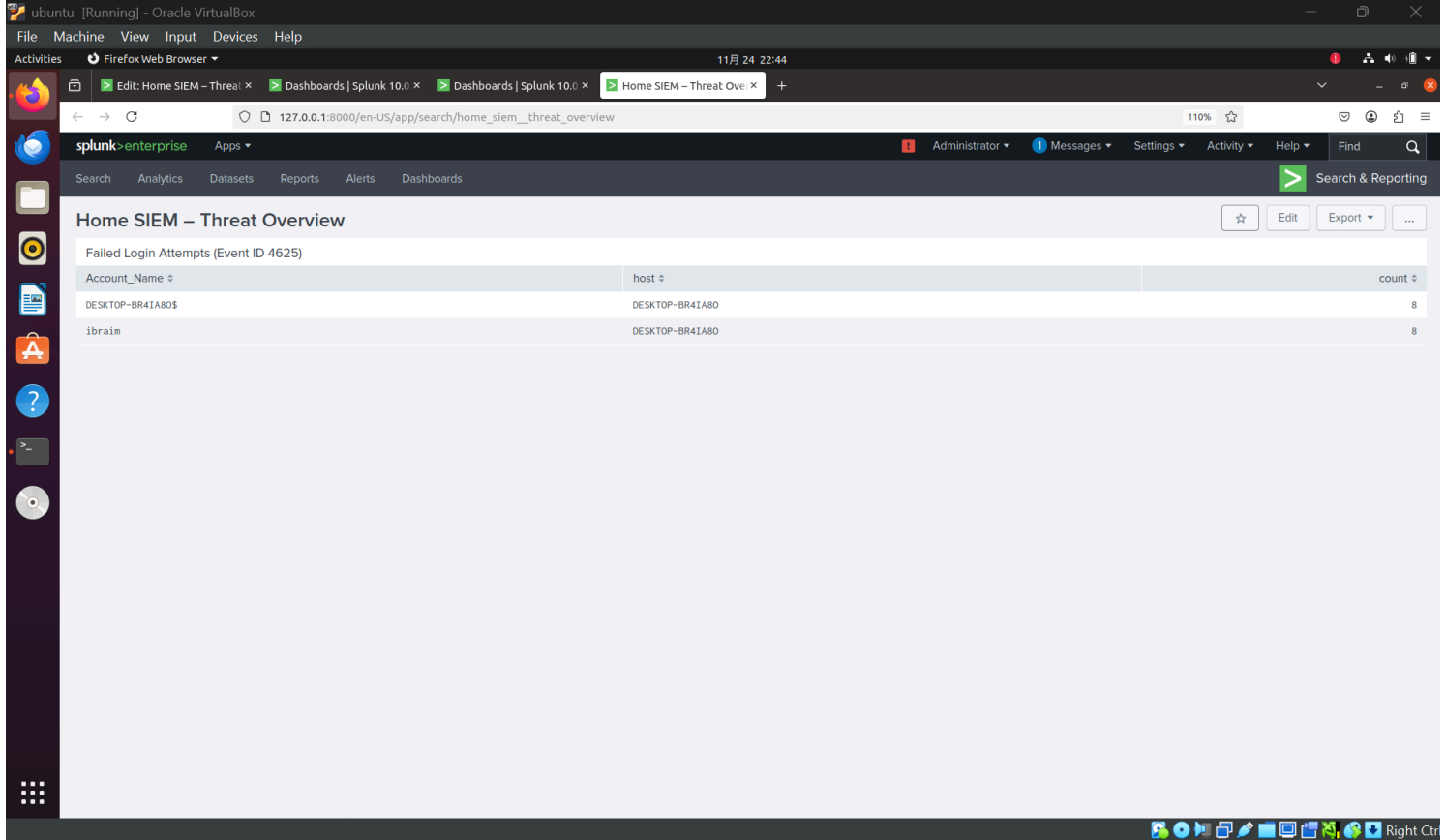
This type of telemetry can help analysts identify registry-based persistence techniques, investigate suspicious system modifications, and improve detection of attacker activity.

8. Home SIEM Dashboard – Threat Overview

After validating detections and attack simulations, I consolidated key security events into a single dashboard to improve visibility and support faster analysis.

The dashboard brings together authentication failures, PowerShell activity, process creation events, and registry modifications, providing a centralized view of security-relevant activity within the environment.

This allows analysts to quickly identify notable events, prioritize investigations, and gain visibility across multiple telemetry sources from a single location.

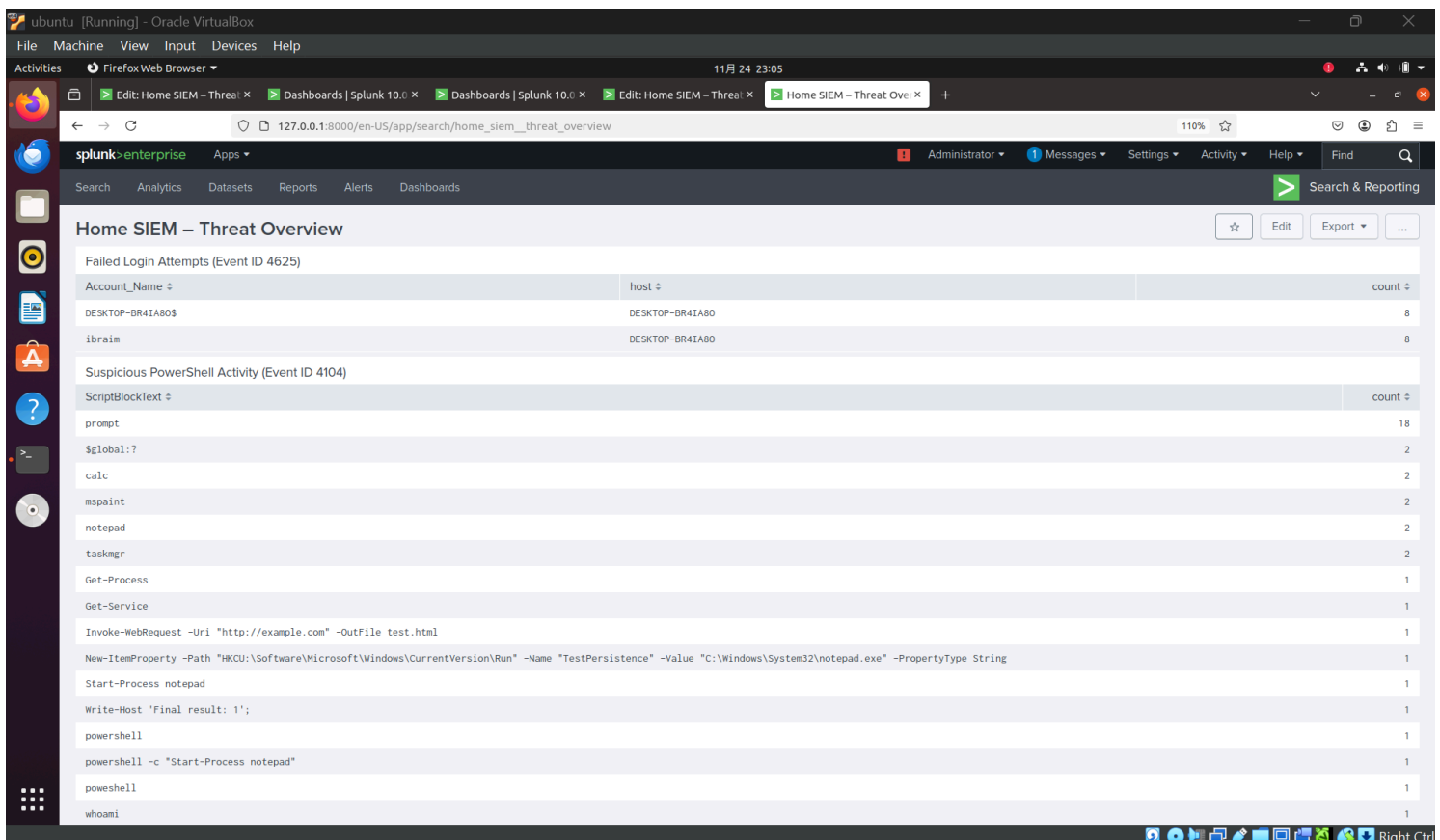


Dashboard Observation

This dashboard panel highlights failed authentication attempts collected from Windows Security Event ID 4625.

By aggregating failed logon events by account and host, the dashboard provides a quick way to identify suspicious authentication activity and prioritize investigation efforts.

8.2 Panel 2 – Suspicious PowerShell Activity (Event ID 4104)



Dashboard Observation

This dashboard panel displays PowerShell Operational Event ID 4104 activity collected from the Windows endpoint.

The panel provides visibility into executed PowerShell commands, allowing both administrative and potentially suspicious script activity to be reviewed from a single dashboard.

By centralizing PowerShell telemetry, analysts can quickly identify unusual command execution patterns, investigate potentially suspicious activity, and gain additional context during security investigations.

8.3 Panel 3 – Process Creation Overview (Sysmon Event ID 1)

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 11月 24 23:18

Dashboards | Splunk 10.0 x Edit: Home SIEM – Threat Ove x Home SIEM – Threat Ove x

127.0.0.1:8000/en-US/app/search/home_siem_threat_overview 110%

{ Set-StrictMode -Version 1; \$this.Exception.InnerException.PSMessageDetails }

1

Process Creation Overview (Sysmon Event ID 1)

Image ↕	CommandLine ↕	count ↕
C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe"	166
C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe"	166
C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	166
C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe"	165
C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	165
C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"	165
C:\Windows\System32\WolsoCoreWorker.exe	C:\Windows\System32\WolsoCoreWorker.exe -Embedding	11
C:\Windows\System32\svchost.exe	C:\Windows\system32\svchost.exe -k netsvcs -p -s gpsvc	11
C:\Windows\System32\taskhostw.exe	taskhostw.exe	9
C:\Windows\System32\wermgr.exe	"C:\Windows\system32\wermgr.exe" -upload	6
C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /ua /installsource scheduler	5
C:\Windows\System32\backgroundTaskHost.exe	"C:\Windows\system32\backgroundTaskHost.exe" -ServerName:App.AppXpskvk16gk8da8kch5g4qx42vxccved.mca	5
C:\Users\ibraim\AppData\Local\Microsoft\OneDrive\25.209.1026.0002\FileCoAuth.exe	"C:\Users\ibraim\AppData\Local\Microsoft\OneDrive\25.209.1026.0002\FileCoAuth.exe" -Embedding	4
C:\Windows\System32\smartscreen.exe	C:\Windows\System32\smartscreen.exe -Embedding	4
C:\Windows\WinSxS\amd64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.19041.6465_none_7e0fb53c7c8be091\TiWorker.exe	C:\Windows\winsxs\amd64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.19041.6465_none_7e0fb53c7c8be091\TiWorker.exe -Embedding	4
C:\Windows\servicing\TrustedInstaller.exe	C:\Windows\servicing\TrustedInstaller.exe	4
C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /c	3
C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /ua /installsource core	3
C:\Program Files (x86)\Microsoft\Edge\Application\142.0.3595.94\elevation_service.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\142.0.3595.94\elevation_service.exe"	3
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"	3

« Prev 1 2 3 4 5 6 7 8 9 10 Next »

Dashboard Observation

This dashboard panel displays Sysmon Event ID 1 process creation activity collected from the Windows endpoint.

The panel provides visibility into executed processes, command-line arguments, and process frequency, allowing endpoint activity to be reviewed from a centralized dashboard.

By monitoring process creation telemetry, analysts can quickly identify unusual process executions, suspicious command-line activity, and abnormal execution patterns that may require further investigation.

This visibility provides valuable context during endpoint investigations and can help analysts understand how processes interact within a system..

8.4 Dashboard Overview (All Panels Combined)

Account_Name	Host	count
DESKTOP-884248D	DESKTOP-884248D	8
ibrahim	DESKTOP-884248D	8

ScriptBlockText	count
prompt	18
\$global:?	2
calc	2
regedit	2
notepad	2
taskmgr	2
Get-Process	1
Get-Service	1
Invoke-WebRequest -Uri "http://example.com" -OutFile test.html	1
New-ItemProperty -Path "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" -Name "TaskPersistence" -Value "C:\Windows\System32\notepad.exe" -PropertyType String	1
Start-Process notepad	1
Write-Host "Final result: 1"	1
powershell	1
powershell -c "Start-Process notepad"	1
powershell	1
whoami	1
(Set-StrictMode -Version 1; \$_.ErrorCategory_Message)	1
(Set-StrictMode -Version 1; \$_.OriginalInfo)	1
(Set-StrictMode -Version 1; \$_.PSMessageDetails)	1
(Set-StrictMode -Version 1; \$this.Exception.InnerException.PSMessageDetails)	1

Image	CommandLine	count
C:\Program Files\SplunkUniversalForwarder\bin\splunk-monitoringd.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-monitoringd.exe"	188
C:\Program Files\SplunkUniversalForwarder\bin\splunk-edmon.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-edmon.exe"	188
C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	188

Dashboard Observation

This dashboard provides a centralized view of the key security events collected throughout the project.

By combining authentication failures, PowerShell activity, process creation telemetry, and registry modifications into a single interface, the dashboard improves visibility and allows analysts to quickly review multiple sources of security telemetry during an investigation.

The dashboard demonstrates how SIEM platforms help analysts monitor multiple data sources simultaneously, prioritize investigative efforts, and identify activity that may require further analysis.

9. Findings

This section summarizes the key findings obtained throughout the Home SIEM project after configuring the log pipeline, creating detections, performing attack simulations, and analyzing security telemetry in Splunk.

Finding 0 – Log Pipeline Successfully Operational

The Home SIEM environment successfully collected and centralized Windows Security Logs, PowerShell Operational logs, and Sysmon telemetry.

Testing confirmed that the log ingestion pipeline was functioning correctly,providing the foundation required for monitoring, detection, and security investigations throughout the project.

Finding 1 – Repeated Failed Authentication Attempts (Event ID 4625)

The investigation confirmed that Windows Security Event ID 4625 telemetry was successfully collected and analyzed through Splunk.

Multiple failed authentication attempts were detected during testing, demonstrating that the SIEM could identify authentication anomalies commonly associated with brute-force activity.

Finding 2 – Suspicious PowerShell Activity (Event ID 4104)

PowerShell Operational Event ID 4104 logs provided visibility into executed commands and script activity on the Windows endpoint.

The collected telemetry demonstrated that both administrative and potentially suspicious PowerShell activity could be monitored, searched, and investigated through centralized logging.

Finding 3 – High Visibility into Process Creation Activity (Sysmon Event ID 1)

Sysmon Event ID 1 telemetry provided detailed visibility into process execution activity on the Windows endpoint.

The collected logs included process names, command-line arguments, parent processes, and user context, supporting endpoint monitoring, threat hunting, and investigation of potentially suspicious activity.

Finding 4 – Registry Modification (Persistence Attempt) Detected (Sysmon Event ID 13)

Sysmon Event ID 13 telemetry successfully captured registry modification activity associated with Windows Run keys.

The collected logs provided visibility into affected registry paths, modified values, and user activity, demonstrating that persistence-related behavior could be monitored and investigated through Splunk.

Finding 5 – Unified Threat Visibility Through the Dashboard

The Home SIEM dashboard successfully consolidated authentication events, PowerShell activity, process creation telemetry, and registry modifications into a centralized monitoring interface.

This unified view improved visibility across multiple data sources and demonstrated how SIEM platforms help analysts correlate events, prioritize investigations, and support security monitoring within a SOC environment.

Summary of Findings

The Home SIEM environment successfully demonstrated end-to-end log collection, threat detection, endpoint visibility, and dashboard-driven monitoring while simulating core SOC investigation workflows through Splunk Enterprise.

Detection Category	Event Source	Result
Authentication failures	Event ID 4625	Detected
Suspicious PowerShell execution	Event ID 4104	Detected
Process creation telemetry	Sysmon Event ID 1	Captured
Persistence modification	Sysmon Event ID 13	Captured
Dashboard correlation	Multiple sources	Operational
Log pipeline integrity	UF + Splunk	Fully Functional

10. Lessons Learned

Throughout this project, I gained practical experience that reinforced several key concepts commonly encountered in SOC operations. These lessons were learned through building the environment, validating telemetry, investigating security events, and developing detections within Splunk.

10.1 Proper Log Ingestion Is Critical for Reliable Detection

One of the most important lessons from this project was that effective detection depends on reliable telemetry collection.

Throughout the project, I learned that even well-designed detection rules provide little value if security events are not being collected and forwarded correctly.

Validating log ingestion became a critical first step before performing investigations, creating detections, or building dashboards.

10.2 Sysmon Greatly Enhances Endpoint Visibility

This project demonstrated how much additional visibility Sysmon provides compared to standard Windows Event Logs.

By collecting process creation events, command-line activity, and registry modifications, Sysmon helped reveal endpoint behavior that would otherwise be difficult to investigate.

The experience reinforced why Sysmon is widely used in enterprise security monitoring and threat detection programs.

10.3 Detection Rules Must Be Tested Through Validation

One of the key lessons from this project was that creating a detection rule is only the beginning. Detection logic must be validated to ensure that alerts are generated as expected and provide meaningful visibility into security events.

By performing controlled attack simulations and reviewing the resulting telemetry in Splunk, I was able to confirm that the detection logic was functioning correctly and generating reliable results for monitoring and investigation.

This reinforced the importance of validating detections before relying on them in a real-world monitoring environment.

10.4 PowerShell Logging Is Essential to Detect Living-off-the-Land Techniques

This project reinforced the importance of PowerShell logging for security monitoring.

By enabling Event ID 4104, I gained visibility into executed commands and script activity, helping distinguish normal administrative behavior from potentially suspicious actions that may require investigation.

The experience demonstrated how PowerShell telemetry can provide valuable context during investigations and support the identification of potentially malicious command execution.

10.5 Effective Dashboards Improve Analyst Efficiency

Building the Threat Overview dashboard demonstrated how centralized visualization can improve security monitoring.

By consolidating multiple data sources into a single view, the dashboard made it easier to identify unusual activity, prioritize investigations, and review security events without manually searching through large volumes of logs.

10.6 Clear Documentation Is Part of Professional SOC Work

One of the key lessons from this project was the importance of documenting investigations, findings, and detection logic in a structured manner.

Clear documentation improves communication, supports collaboration, and helps ensure that investigations can be understood and reviewed by other analysts.

The experience reinforced the importance of maintaining clear records throughout an investigation to support knowledge sharing and future analysis.

10.7 Endpoint Behavior Is Predictable—Anomalies Are Detectable

By observing normal endpoint activity throughout the project, it became easier to recognize unusual behavior during testing and investigations.

This experience reinforced the importance of understanding baseline activity, as identifying deviations from normal behavior is a fundamental part of threat detection and SOC analysis.

Establishing a baseline of normal activity provides valuable context that helps analysts identify potential security events more effectively.

11. Future Improvements

The current Home SIEM environment provides foundational visibility into authentication events, PowerShell activity, process creation, and registry modifications.

While the project successfully demonstrated core SIEM monitoring and investigation workflows, several enhancements could further improve detection coverage, scalability, automation, and alignment with enterprise SOC practices.

11.1 Expand Sysmon Configuration for Deeper Coverage

While the current Sysmon configuration provided valuable visibility into process creation and registry activity, future improvements could expand monitoring coverage across additional attack techniques.

Adding telemetry related to network connections, file modifications, image loading, and remote thread creation would provide deeper visibility into attacker behavior and improve overall detection capability.

11.2 Add Additional Log Sources Beyond the Single Windows Endpoint

This project focused on a single Windows endpoint to validate core SIEM concepts. A future enhancement would be integrating additional log sources such as Linux systems, network devices, security tools, and endpoint protection platforms.

Expanding data collection across multiple sources would enable more realistic correlation, broader visibility, and stronger detection capabilities similar to those found in enterprise environments.

11.3 Implement Scheduled Alerts and Notifications

Most detections within this project were validated through manual investigation and dashboard review.

A future improvement would be implementing automated alerting and notification workflows to support faster detection and response, bringing the environment closer to a real SOC monitoring workflow.

11.4 Expand MITRE ATT&CK Mapping Across Future Detections

Although several detections were mapped to MITRE ATT&CK techniques, future improvements could expand coverage across additional tactics, techniques, and procedures.

Broader ATT&CK mapping would strengthen detection context, improve investigation workflows, and provide a more comprehensive view of adversary behavior.

11.5 Add Threat Hunting Queries and Playbooks

A future enhancement would be developing dedicated threat hunting queries and investigation playbooks to support more structured analysis workflows.

This would help standardize investigations, improve consistency, and simulate analyst activities commonly performed within mature SOC environments.

11.6 Implement Data Normalization and CIM Compliance

Future improvements could include normalizing log data using Splunk's Common Information Model (CIM).

Applying standardized data models would simplify correlation, improve dashboard functionality, and support more advanced security monitoring use cases.

11.7 Expand Dashboard Functionality

While the current dashboard provides useful visibility into security events, future versions could incorporate additional visualizations, filtering options, and trend analysis capabilities.

These improvements would enhance the analyst experience and provide deeper insight into endpoint activity over time.

11.8 Automate Environment Deployment

Automating the deployment of the Home SIEM environment would improve consistency, scalability, and reproducibility.

Implementing infrastructure and configuration automation would make it easier to rebuild, maintain, and share the environment while following modern operational practices.

11.9 SOC Analyst Skills Demonstrated

Throughout this project, I developed and demonstrated practical SOC Analyst skills, including:

- Log collection and analysis
- Security event investigation

- Detection rule development and validation
- Threat simulation and testing
- PowerShell and Sysmon analysis
- Dashboard creation and monitoring
- Incident documentation and reporting
- Security monitoring using Splunk Enterprise
- MITRE ATT&CK mapping
- Security telemetry analysis

12. Conclusion

This project provided practical experience with the end-to-end workflow commonly performed by SOC Analysts.

Starting from a basic lab environment, I built a centralized logging solution using Splunk Enterprise, collected security telemetry from a Windows endpoint, developed detection rules, validated those detections through controlled attack simulations, and investigated the resulting events.

Throughout the project, I gained hands-on experience reviewing authentication events, PowerShell activity, process creation telemetry, and registry modifications. These investigations reinforced the importance of reliable log collection, endpoint visibility, detection validation, and structured documentation.

Beyond the technical implementation, the project strengthened my ability to think analytically, investigate security events, validate hypotheses, and communicate findings in a structured manner. It also provided valuable insight into how SIEM platforms support monitoring, detection, investigation, and response activities within a SOC environment.

This Home SIEM project provides a strong foundation for future growth in threat hunting, incident response, detection engineering, and Security Operations Center (SOC) roles.

13. Appendix – Technical References & Commands

This appendix contains the technical references used throughout the project, including SPL queries, attack simulation commands, and supporting configurations.

The purpose of this section is to provide transparency into the investigative methodology, support reproducibility, and allow the project findings to be validated.

A. SPL Queries Used in the Project

1. Failed Login Attempts (Event ID 4625)

```
index=main source="WinEventLog:Security" EventCode=4625
| stats count as failed_attempts by Account_Name, host
| where failed_attempts >= 5
```

2. Suspicious PowerShell Execution (Event ID 4104)

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-PowerShell/Operational" EventID=4104
| table _time host ScriptBlockText
| sort -_time
```

3. Sysmon Process Creation (Event ID 1)

```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventID=1
| table _time host Image CommandLine User
| sort -_time
```

4. Registry Modification Detection (Event ID 13)


```
index=main sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventID=13
| table _time host TargetObject Details User
| sort - _time
```

B. Windows Commands Used for Attack Simulations

1. Brute-Force Simulation (Failed Logins)

Multiple failed login attempts were intentionally generated on the Windows endpoint by entering incorrect credentials several times.

This activity generated Windows Security Event ID 4625 telemetry used for validation and investigation.

2. PowerShell Simulation Commands

```
Get-Process
Get-Service
whoami
Invoke-WebRequest -Uri "http://example.com" -OutFile test.html
New-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" -Name "TestPersistence" -
Value "C:\Windows\System32\notepad.exe" -PropertyType String
Start-Process notepad
```

3. Process Creation Simulation

```
notepad
calc
taskmgr
powershell
```

C. Sysmon Configuration (Used in the Project)

The Sysmon configuration used throughout the project focused on capturing endpoint telemetry required to support monitoring, investigations, and attack simulations.

Key event types included Process Creation (Event ID 1) and Registry Modification (Event ID 13), providing visibility into process execution activity and persistence-related behavior.

```
Sysmon64.exe -accepteula -i sysmonconfig.xml
```

D. Splunk Universal Forwarder Verification

The Splunk Universal Forwarder service was verified to ensure endpoint telemetry was being collected and forwarded successfully to Splunk Enterprise.

Windows Service Verification

```
services.msc
```

PowerShell Verification

```
Get-Service SplunkForwarder
```

E. Splunk Enterprise Access and Management

Start Splunk (Ubuntu)

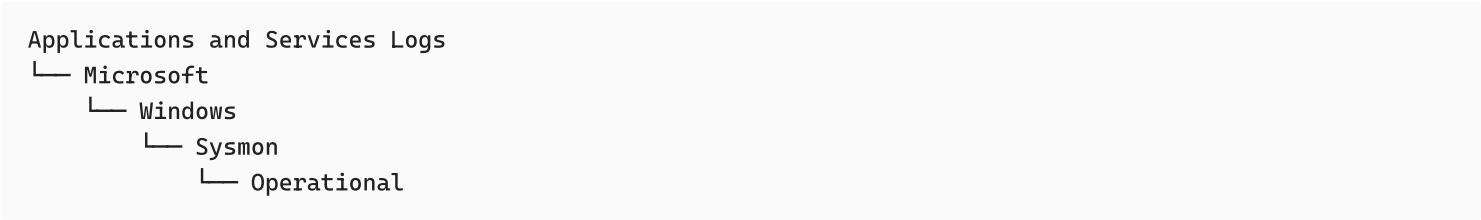
```
sudo /opt/splunk/bin/splunk start
```

Access Splunk Web

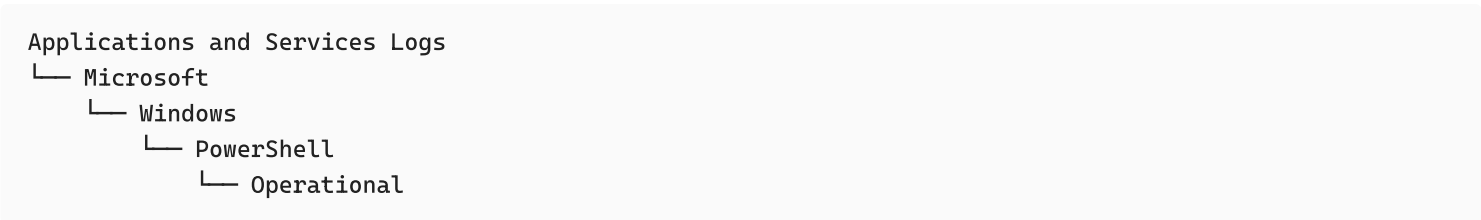
```
http://127.0.0.1:8000
```

F. Important Paths

Sysmon Logs



PowerShell Script Block Logs



Security Logs



G. Dashboard Panels (as Implemented)

Failed Login Attempts
Windows Security Event ID 4625

Suspicious PowerShell Activity
PowerShell Operational Event ID 4104

Process Creation Overview
Sysmon Event ID 1

Registry Persistence Activity
Sysmon Event ID 13